tainly don't think computers should have to be able to come up with a unified theory of gravity to be considered "intelligent", but we do want a little more than the Turing test offers. Surely something that is given the title of "intelligence", artificial or otherwise, will have to be more than a mere charlatan!

Consider Deep Blue beating Kasparov at chess, or Watson winning at Jeopardy, or Google DeepMind's AlphaGo beating Lee Sedol and Ke Jie at Go... these are examples of computers beating humanity's best and brightest at their own game! Should that be considered AI? Many would say yes, we still think it's not quite the same. AlphaGo probably sucks at poker, and Deep Blue would flunk at backgammon. Of course they could be coded and re-built to play those games, but it would take years. And even after that, a 6 year old could ask them questions that would stump them. So are they AI? We think not.

Then there are those who will claim that a computer has to be sentient, be self aware, and be able to learn, before it can considered intelligent. In short, they want the birth of artificial life, which matures over time, much in the way we change from when we are babies to when we become thinking adults. Obviously, because such an AI would literally mimic human development, it would have to be considered "intelligent".

Positronic Brains?

Is that what we're after? The imagination of Isaac Asimov ruling our future still? What we mean is, are we looking to create artificial humans? Because the sample set of intelligent beings we know of is one, much like the sample set of known planets with life, are we just too ignorant to spot intelligence when we see it?

From what we know, an ant is essentially a robot that needs to eat and uses primitive senses to do so. It has a few laws inside it that define it. For example, it defends itself, or even flees when it is threatened. It communicates, it undertakes the task of searching for food, of protecting the nest, of sacrificing itself for its queen, etc. Are ants intelligent? No relativity needed, not are they intelligent as humans, or dolphins... just, "are they intelligent"?

Depending on whether or not you think of an ant as intelligent (or a bee, or a fish...) will help you best understand what you consider to be intelligence, and will help you answer the question we've posed.

Do we already have AI?

If you answered "Yes, ants are intelligent", then yes, we already have AIs that intelligent. The rovers on Mars are like huge ants that are going about their business, with no hookup to the internet, and no real-time contact with anything. They're autonomous, and are crawling about on Mars looking for stuff that's interesting. Of course we still command them, much like an ant will report back to the colony and then be told to go do something. Both the rovers and the ants have "masters" who essentially control them. Heck, you and I have bosses who tell us what to do, and we are all expendable when it comes to work. Don't believe me? The next time your boss tells you to do something, tell him to go to hell; then see what happens.

Asimov's AI is what we're trying to make

The mistake a lot of us make is to only consider a human standard of intelligence. An ant doesn't need art or social media to feel good about itself, and for all we know, ants don't even have the concept of feeling good. Similarly, expecting an AI to behave like a human is probably a mistake. We're not mathematically driven objects – one look at the average results of a math exam will tell you that! We have abstract thought, creativity, depression, happiness, anger, and all of that non-mathematical and illogical input makes us who we are. How can we build something based purely on mathematics and then expect it to behave like us?

What we're getting at is that if we give up this notion that human-like = intelligence, we might realise that we have artificial intelligence all around us. It's task specific and hasn't mastered the nuances of the languages we use yet, but it is precise and predictable and dependable. How often can you say that about humans?

We already know that computers will take over driving cars, and do it better, with less accidents than humans. Of course driving will be reduced to commuting if that happens, and no one is going to enjoy it anymore than they enjoy riding the train to work, but it will happen eventually anyway. Are Tesla's cars AI?

Smart assistants have a long way to go still, but then again, the first artificial driving assist couldn't drive a car either. Are Siri, Alexa, Cortana and GA (Google Assistant) just toddlers in a pre-kindergarten program? After all, babies are dumb too, but they have milestones that they're supposed to reach in order to be considered of average intelligence... So is current AI like a toddler, or a baby, and what are the milestones we should set for it? We can't set time frames the way we can with human children – by 24 months they should say words, etc. However, can we still set milestones – when an AI is able to learn new words, is it a toddler, when it can drive a car, is it 16?

Maybe the stories that follow can help you decide whether we already have AI or not...